a. Find: the delay creeps, then jumps one frame
ECG to PPG (ms)
760
740
720
700
b. Measure: 8 slips; the sawtooth fitted (black, 44.8 ppm)
less its slow trend (ms)
20
0
−20
c. Undo: corrected (orange); what remains is physiology
ECG to PPG (ms)
760
740
720
700
0
20
40
60
80
minutes (recording 5328)
d. Test: steps at 663 slips (265 recordings)
corrected median −0.5 ms
measured median 31.9 ms
slips
200
100
0
−20
0
20
40
step at each slip (ms)
e. Test: the arterial line (recording 5328)
15-beat medians
150
100
50
0
0
100
200
300
400
500
600
700
800
delay from the R peak (ms)
ECG to arterial (robust SD 3.5 ms)
ECG to PPG, corrected (robust SD 7.6 ms)
ECG to PPG, as recorded (robust SD 15.1 ms)